

5.2 Commissioning data

5.2.1 Option parameters

The PPC6000 series is configured by programming Option Parameters (memory registers) that describe the configuration of the burner and the boiler. Option parameters are set and adjusted in *Option Set* mode, but it may not be possible to adjust all of them if the burner is on.

5.2.2 Setpoints

Setpoints contain information about required motor positions. There are four profiles (or tables) of setpoints available in a standard control. The profiles may be represented using the diagram below:

Gas position (°)	Air position (°)		Oil position (°)	Air position (°)
2.1	1.9	CLOSE (P0)	1.7	1.9
2.1	88.9	PURGE (P1)	1.7	88.9
24.6	30.6	IGNITION (P2)	10.6	28.7
21.3	25.8	LOW FIRE (P3)	10.6	28.7
76.8	85.6	HIGH FIRE (Px)	50.2	83.5

Up to 24 setpoints may be entered for each profile, including close, purge, and ignition.

New setpoints must be entered in *commission ratio mode*.

Existing setpoints may be modified in *adjust ratio mode* or *commission ratio mode*.

5.3 The Commissioning process

5.3.1 Navigation

The symbols below will be used throughout the Commissioning Process in this manual to indicate the correct application for each commissioning OPTION, for use by the Commissioning Personnel.

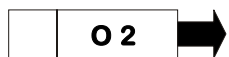
A convenient check box has been added to the beginning of the symbols so that the Commissioning Personnel can mark each appropriate option as it is completed.

SYMBOL

MEANS:



Commissioning Personnel **MUST** execute this option



For Customers with **O2 TRIM** option, Commissioning Personnel **MUST ALSO** execute this option



For Customers with **VSD** options, Commissioning Personnel **MUST ALSO** execute this option



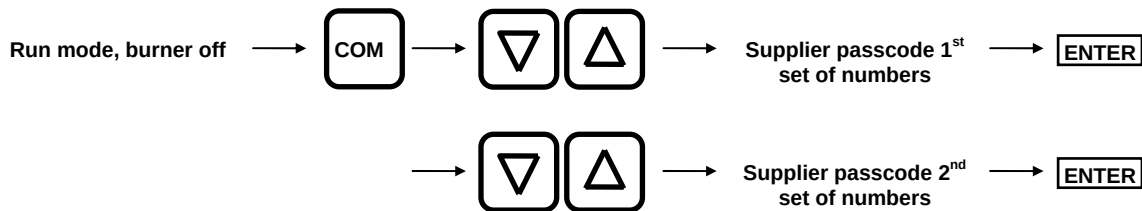
After each appropriate option parameter is complete, **CHECK THE BOX!**



There are two (2) types of Displays available for the PPC6000. The first is the standard 12-key keypad shown below and the second is the optional 10.4" touchscreen (NXTSD104) described in section 5.4. Both displays rely on the information in section 4, and the manual reads as though the standard display is being used, so please read the entire section 4 regardless of the type of display in the configuration being used.

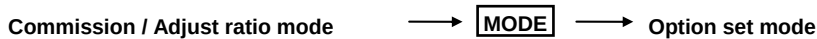
5.3.1.1 Entering commission ratio mode

To enter commission ratio mode, turn the burner off. Once the burner has completely shut down, press the key labeled **COM** and select the supplier passcode value using the **UP/DOWN** keys. The supplier passcode is normally two sets of three numbers and is entered as shown below.



If the entered passcode was correct, 'P 0' will be seen on the display. This represents the closed position setpoint.

5.3.1.2 Changing from Commission / Adjust Ratio mode to Option Set mode



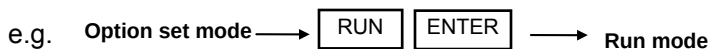
To enter option set mode, enter commission ratio mode or adjust ratio mode (see sections 5.3.1 or 5.4.14 respectively) and press the key labeled **MODE**.

5.3.1.3 Changing from option set to Commission / Adjust ratio modes



5.3.1.4 Exiting from commission mode

To exit from any commission mode to run condition, press the key labeled **RUN** and then **ENTER**.



NOTE: Exiting from "full commissioning" mode at P3 or higher will erase any point above the displayed Px. If this is **undesirable**, exit "full commissioning" mode at **HIGH FIRE**.

5.4 Description of Touchscreen operation.

5.4.1 The Touchscreen

The touchscreen provides all of the functions required to control and monitor the burner, it is also used to commission the control.

Once commissioning mode is selected the display background changes to red to alert the user that the control is in commissioning mode and therefore unable to monitor certain safety functions such as fuel air ratio positions.

For a detailed overview of the NXTSD104 – see bulletin NXOGTSD-6101 Operator's Guide.

5.4.2 Touchscreen Power on.

Initialization of the control and display will take approximately 16 seconds, during this initialization period the status LED will remain red. The display backlight will come on and the relays will be held in their no alarm state to avoid the possibility of nuisance alarms, which may otherwise occur.

Five (5) seconds after power is applied the Fireeye splash screen (shown below) will be displayed for 6 seconds, after which a period of 5 seconds of a blank white screen will show.

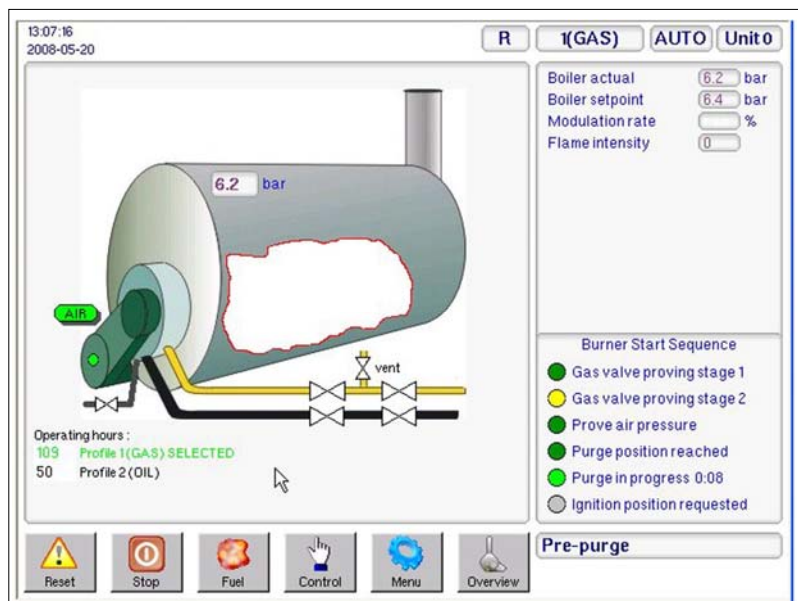


Once the initialization is complete the touchscreen will show the overview screen (shown in section 5.4.3) and the control will operate normally, changing the LED to green and relays to operate in line with the control status.

5.4.3 The Operation Mode line and Status messages

The line at the top of the screen displays the current burner: operating mode, profile, modulation mode and unit address. Burner status, and fault information, if applicable, is displayed in the bottom right corner of the display.

Dependent on the actual fault the audible alarm may be active and the control may move to safety shutdown.



Dedicated 'buttons' at the lower edge of the screen perform the following functions:

Mute / Reset – used to reset a burner lockout, mute alarms or to test audible alarms. If an alarm is present the button is yellow and the text on the 'button' changes to Alarm Mute.

Burner Start/Stop – starts the burner, grayed out if the burner is already operating.

Fuel – allows selection of which fuel/profile is to be used, from a list of fuels/profiles available.

Control – used to enable selection of modulation mode.

Menu – used to access configuration, adjustment modes.

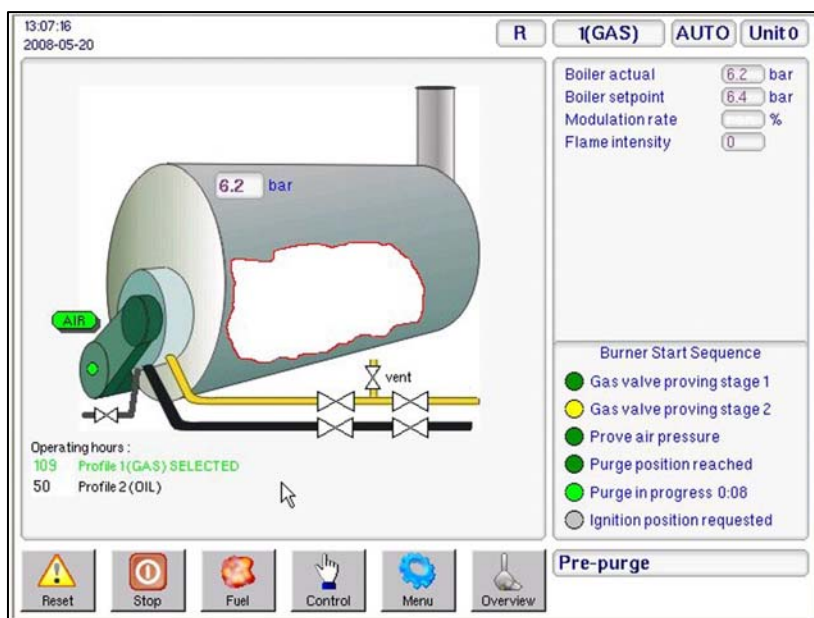
Overview – returns screen to burner overview mode.

The following table shows the Icons that are displayed on the touchscreen above the information panels and their details.

Icon	Description	Details
	State of CANbus connection	Displays when CANbus communication is faulty
	State of Oxygen Interface	Displays when Oxygen Interface related fault is present.

Icon	Description	Details
COM	Indicates commissioning mode	Displays when the control is operating in commissioning mode.
MAN	Indicates HAND modulation mode.	Displays when the control is operating in HAND modulation mode.
AUTO	Indicates AUTO modulation mode.	Displays when the control is operating in AUTO modulation mode.
R / L(1) / L(2)	Indicates Boiler Setpoint mode	Displays Boiler setpoint selected, remote, local setpoint 1 or local setpoint 2.
1 / 2 / 3 / 4	Indicates profile selected	Displays the profile number for the selected profile.
13:07:16 2008-05-20	Indicates Time and Date	Displays Actual Time and Date, used for fault history and event logging. Date format is yyyy-mm-dd

5.4.4 Touchscreen Overview display.



While the burner is changing 'status' the bottom-right one sixth of the display is used to provide a graphical indication of changing status.

The sequences displayed in this manner include, burner start-up, burner shutdown and fuel changeover.

In each case there is a 'lamp' for each stage, providing a ripple through lamp system to clearly indicate the progress of the change in burner 'status'.

The example shown in the picture is for the burner start-up sequence display.

Once the sequence is complete then the display will return, after a short delay, to the overview screen.



5.4.5 Touchscreen Lockout Reset (Alarm Mute) Function.

The **Reset** (Alarm Mute) 'button' on the touchscreen is multi-functional. In normal operation it may be used to test the alarms, this is achieved by holding the button for in excess of 10 seconds after which the alarm relays will activate.

The **Reset** (Alarm Mute) button is also used to mute alarms and reset the control following a safety shutdown. If the alarm is sounding, the button will show the text **Mute**, pressing the button will mute (silence) the alarm, and then the text in the button will change to **Reset**, holding the button for greater than 3 seconds will allow the burner to restart if the faults have cleared. Further details of the functionality of this button are defined below.

There are three types of fault (alarm):

Alarm Only This will allow the burner to continue to operate, while the alarm is sounding.

If the **Reset** (Alarm Mute) button is pressed while the control is in this mode and the alarm is sounding the alarm will be muted and the burner will continue to operate.

Controlled Shutdown This will cause the burner to perform a controlled shutdown, and the alarm will sound, once the fault clears the burner will restart without the requirement for manual intervention.

If the **Mute/ Reset** button is pressed while the control is in this mode the alarm will be muted and the burner will remain in controlled shutdown.

If the **Mute/Reset** button is pressed after the control has restarted following a controlled shutdown the alarm will be muted and the burner will continue to operate.

Safety Shutdown This will cause the burner to perform a safety shutdown, and the alarm will sound, once the fault has cleared the burner will remain in safety shutdown until a **Mute/ Reset** is performed.

If the **Mute/ Reset** button is pressed while the control is in safety shutdown and the fault is still present the alarm will be muted and the burner will remain in safety shutdown. When the fault clears the alarm will sound once more.

If the **Mute/ Reset** button is pressed while the control is in safety shutdown and the fault has cleared the alarm will be muted and if the button is held for in excess of 3 seconds the burner will re-start.

5.4.6 Touchscreen LED Indication.

The Touchscreen also houses a multi-functional LED for indication of current status.

The LED will be green, permanently on, if no faults or limits are present.

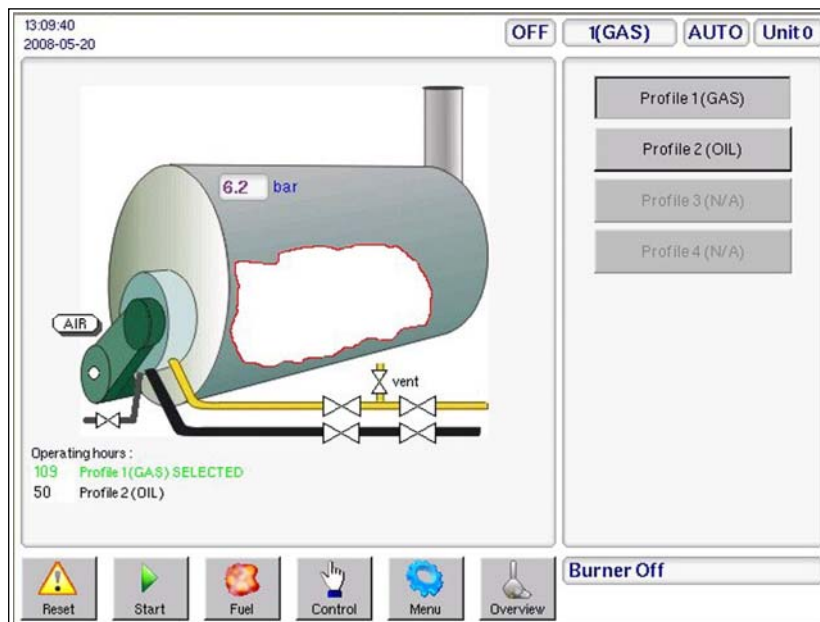
The LED will be red, flashing, if there is a limit or an alarm condition present, which does not cause a lockout.

The LED will be red, permanently on, if there is a fault present, even if the fault has been muted, which causes a lockout.

5.4.7 Touchscreen Fuel/Profile Select.

Once the **Fuel** soft-button is pressed, a pop-up window appears on the right giving the user a choice of available fuels/profiles.

The PPC6000 allows for a maximum of four profiles. Any profiles, which are not programmed and therefore not available, will be 'grayed-out'.



The 'Name' for each available profile will be either the default as set by the relevant option parameter or that entered via the Configuration Screen.

If the profile change is made while the burner is firing, there are two possibilities dependent on the options set in the control.

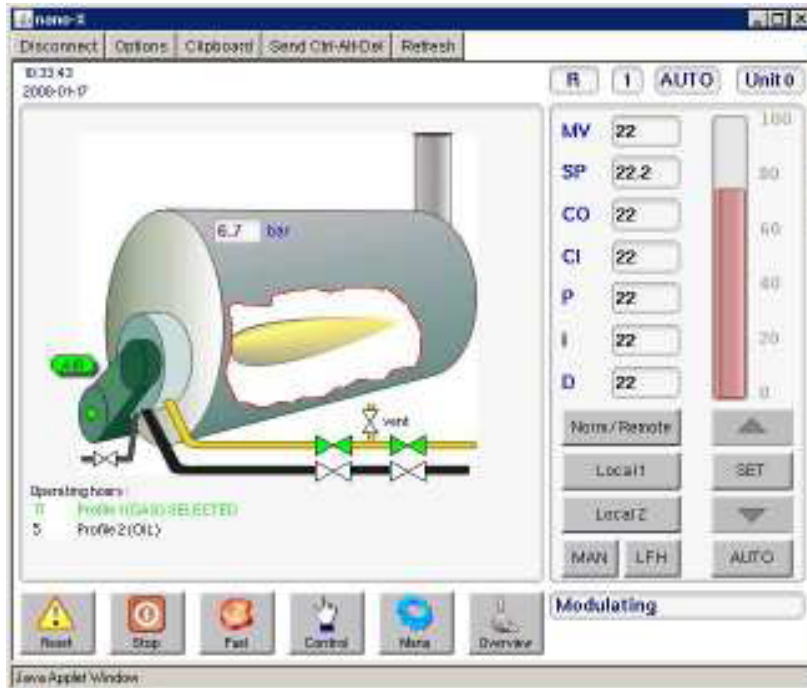
If the unit does not have the option to perform an online changeover enabled (switch profile without the burner going off) the burner will go through a controlled shutdown on the original profile and then restart on the new profile.

The online changeover allows a digital input to be configured to allow a profile swap without turning the burner off. If this input is ON and a fuel profile selection change is made, the control will go to low fire then back to pilot ignition (P2) on the original profile. It will then drop the main fuel valves and run with just the pilot on (and the ignition transformer if option parameter 14.6 is not set to 1). It will then move all drives to the P2 position of the new profile and open the appropriate main fuel valves.

To exit this operation window and return to the overview screen press the **Overview** button.

5.4.8 Touchscreen 'Control' screen.

Once the **Control** button is pressed, a window appears giving the user a choice of options. If an option is unavailable it will be 'grayed out', for example Manual Modulation may be disabled during commissioning.



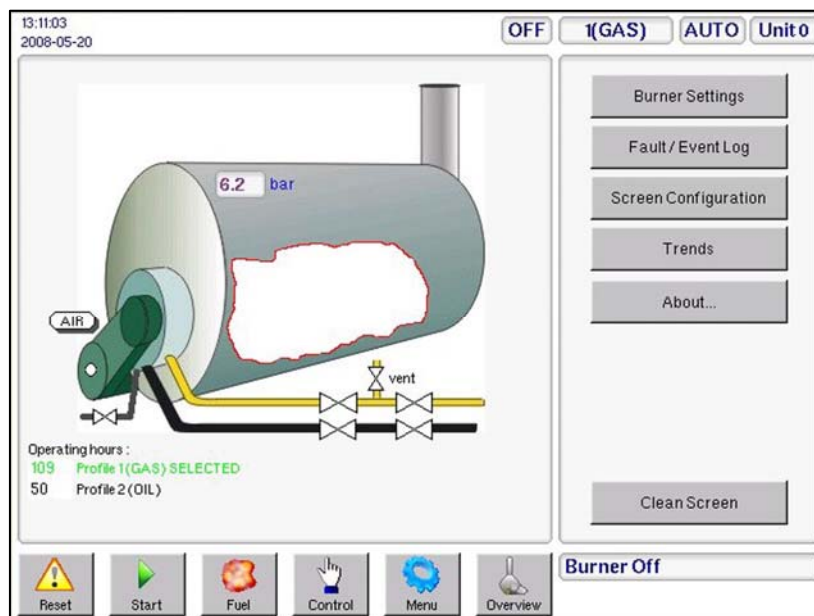
If the burner is firing, then the modulation percentage will be displayed in a window between the 'Burner off' and 'Auto' boxes.

Using the buttons available from this window will change the way the burner is operating.

To exit this control mode window and return to the overview screen use the **Overview** button.

5.4.9 Touchscreen 'Menu' button.

Once the **Menu** button is pressed, a pop-up window appears on the right giving the user a choice of options. If an option is unavailable it will be 'grayed out'.



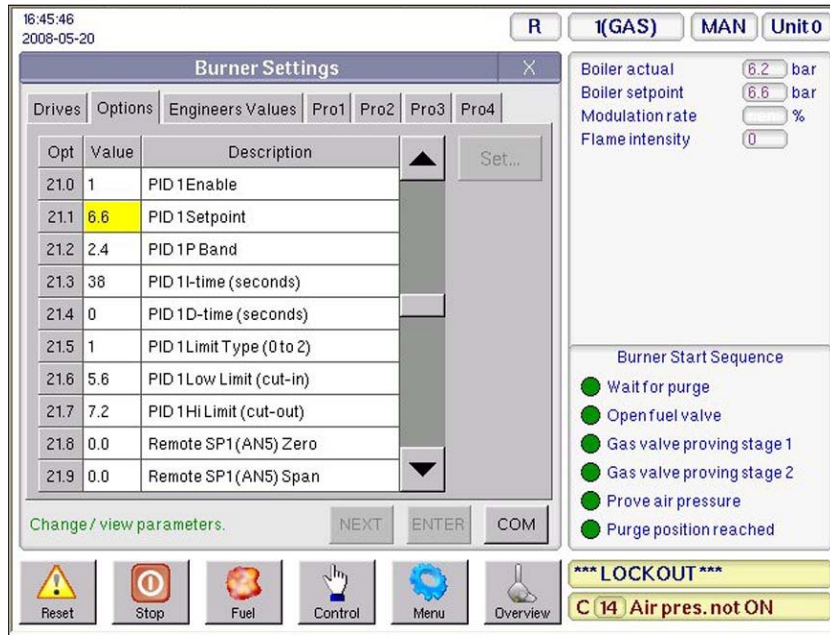
Using the buttons available after pressing the **Menu** button will generate a new pop-up window in the left hand area of the screen.

To exit this operation window and return to the overview screen press the **Overview** button.

5.4.10 Touchscreen 'Burner

Settings' button.

Once the **Burner Settings** button is pressed, a window appears giving the user a choice of data types by Tab. Selecting, by touching a tab, it is possible to view a variety of data, in the example here the Option Parameters are available.



16:45:46
2008-05-20

R 1(GAS) MAN Unit 0

Burner Settings

Drives Options Engineers Values Pro1 Pro2 Pro3 Pro4

Opt	Value	Description
21.0	1	PID 1 Enable
21.1	6.6	PID 1 Setpoint
21.2	2.4	PID 1 P Band
21.3	38	PID 1 I-time (seconds)
21.4	0	PID 1 D-time (seconds)
21.5	1	PID 1 Limit Type (0 to 2)
21.6	5.6	PID 1 Low Limit (cut-in)
21.7	7.2	PID 1 Hi Limit (cut-out)
21.8	0.0	Remote SP1 (AN5) Zero
21.9	0.0	Remote SP1 (AN5) Span

Boiler actual 6.2 bar
Boiler setpoint 6.6 bar
Modulation rate %
Flame intensity 0

Burner Start Sequence

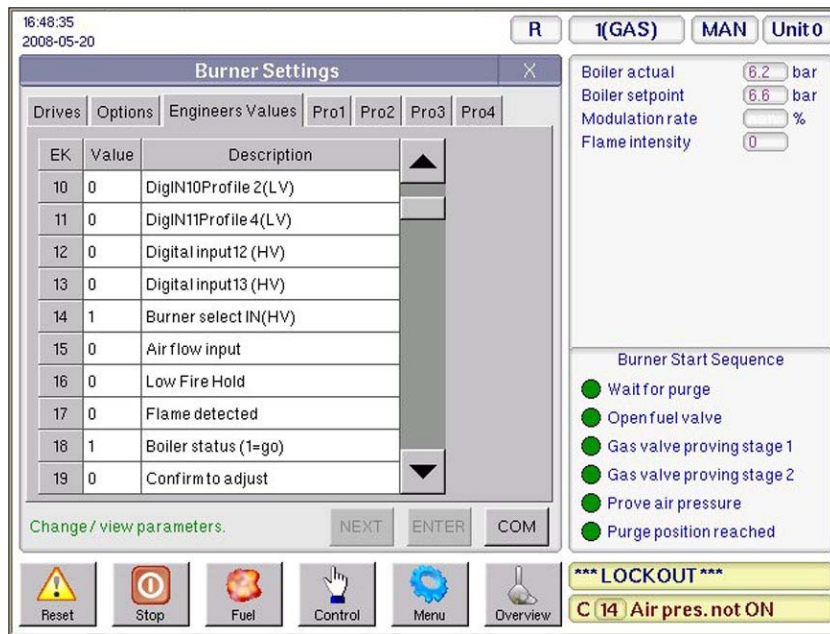
- Wait for purge
- Open fuel valve
- Gas valve proving stage 1
- Gas valve proving stage 2
- Prove air pressure
- Purge position reached

Change / view parameters. NEXT ENTER COM

Reset Stop Fuel Control Menu Overview

*** LOCKOUT ***
C 14 Air pres. not ON

In the following example the Engineers Key values are displayed. The scroll bar at the side of the data tab allows the user to move up and down the table.



16:48:35
2008-05-20

R 1(GAS) MAN Unit 0

Burner Settings

Drives Options Engineers Values Pro1 Pro2 Pro3 Pro4

EK	Value	Description
10	0	DigIn10Profile 2(LV)
11	0	DigIn11Profile 4(LV)
12	0	Digital input12 (HV)
13	0	Digital input13 (HV)
14	1	Burner select IN(HV)
15	0	Air flow input
16	0	Low Fire Hold
17	0	Flame detected
18	1	Boiler status (1=go)
19	0	Confirm to adjust

Boiler actual 6.2 bar
Boiler setpoint 6.6 bar
Modulation rate %
Flame intensity 0

Burner Start Sequence

- Wait for purge
- Open fuel valve
- Gas valve proving stage 1
- Gas valve proving stage 2
- Prove air pressure
- Purge position reached

Change / view parameters. NEXT ENTER COM

Reset Stop Fuel Control Menu Overview

*** LOCKOUT ***
C 14 Air pres. not ON

5.4.11 Touchscreen 'Fault/Event Log' button.

Once the **Fault/Event Log** button is pressed, a window appears showing the Fault/Event history. Users may select to see Fault data only, Events data only or a combined history as shown in this example. For more information on fault codes and fault finding refer to the Fault Finding section of the PPC6000 manual – Fireeye bulletin NEX-6101.



Item	Date/Time	Description	Sbst	Pro	SP
info	2008-05-20 16:51:19	Enter COMMISSION mode	---	1	---
info	2008-05-20 16:50:45	Burner shutdown	---	1	---
info	2008-05-20 16:50:45	Switched OFF	---	1	---
info	2008-05-20 16:50:08	Burner startup	---	1	---
info	2008-05-20 16:50:04	Re-start (Mute)	0	1	3
C14	2008-05-20 15:10:50	Air pres. not ON	0	1	3
F14	2008-05-20 15:10:49	Air pres. not ON	16	1	3
info	2008-05-20 14:23:40	Burner startup	---	1	---
info	2008-05-20 14:23:36	Switched ON (REMOTE)	---	1	---
info	2008-05-20 14:23:30	Burner shutdown	---	1	---
info	2008-05-20 14:23:29	Switched OFF	---	1	---
info	2008-05-20 14:23:27	Switched ON (REMOTE)	---	1	---

5.4.12 Touchscreen 'Screen Configuration' button.

Once the Screen **Configuration** button is pressed, a window appears giving the user a choice of options. From the General Tab a name may be entered for each profile in use.

